Fiscal Year:	FY 2025	**Task Last Updated:**	FY 08/05/2024
PI Name:	Macaulay, Timothy Ph.D.		
Project Title:	Countermeasures for Mitigation of Sensorimotor Decrements Following Head-Down Bed Rest		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **HHC**:Human Health Countermeasures		
Human Research Program Risks:	(1) **Sensorimotor**:Risk of Altered Sensorimotor/Vestibular Function Impacting Critical Mission Tasks		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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City:	Houston	**State:**	TX
Zip Code:	77058-3711	**Congressional District:**	36
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	Directed Research
Start Date:	10/07/2021	**End Date:**	10/06/2027
No. of Post Docs:	1	**No. of PhD Degrees:**	1
No. of PhD Candidates:	1	**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA JSC
Contact Monitor:	Brocato, Becky	**Contact Phone:**	
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Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:	Updates were made to the investigator team, including the additions of Drs. Brandon Macias and Steven Laurie from the NASA JSC Cardiovascular and Vision Laboratory and their external partners Drs. Alex Huang and Larry Kramer. These co-investigators have led the study augmentation of measures for SANS and IJV flow. They will be responsible for the data collection, processing, and analysis of these measures. In addition, Dr. Aenor Sawyer has been added as a co-investigator to provide subject matter expertise on Radiofrequency Echographic Multi Spectrometry (REMS) technology and bone ultrasound assessments. Dr. Sawyer has supported hardware setup and training and will lead the processing and analysis of bone ultrasound measurements.		
COI Name (Institution):	Bent, Leah Ph.D. (University of Guelph, Guelph, Ontario, Canada) Young, Millennia Ph.D. (NASA Johnson Space Center) Oddsson, Lars Ph.D. (University of Minnesota, Minneapolis) Peters, Brian Ph.D. (KBR/NASA Johnson Space Center) Wood, Scott Ph.D. (NASA Johnson Space Center) Rutkove, Seward (Beth Israel Deaconess Medical Center) Mortreux, Marie (Beth Israel Deaconess Medical Center) Beheshti, Afshin (KBR/NASA Ames Research Center) Bouxsein, Mary (Beth Israel Deaconess Medical Center) Valero-Cuevas, Francisco (Neuromuscular Dynamics LLC) Strock, Nicole (KBR/NASA Johnson Space Center)		

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Grant/Contract No.:	Directed Research
Performance Goal No.:	
Performance Goal Text:	
Task Description:	Note: Research aims merged with Grant #80NSSC21K0369 (PI: Rutkove). Astronauts experience postflight disturbances in postural and locomotor control due to sensorimotor adaptations during spaceflight. These alterations may have adverse consequences if a rapid egress is required after landing. Although exercise is effective for mitigating cardiovascular and muscular deconditioning, new approaches are needed to further preserve sensorimotor function, especially during and after exploration missions with less comprehensive exercise capabilities than the current suite on the International Space Station (ISS). We have identified proprioceptive training and electrical muscle stimulation (EMS) as promising in-flight countermeasures. Since prolonged head down bed rest (HDBR) is a spaceflight analog for body unloading and causes postural and locomotor control decrements that parallel those observed after spaceflight, it can be used to accelerate the development of these countermeasures. This study will determine the effects of proprioceptive training and EMS on functional task performance and sensorimotor function following 60 days of 6° HDBR. We will compare four groups of subjects – an EMS arm, a proprioceptive training arm, an exercise plus proprioceptive training arm, and a control arm. The EMS countermeasure will include daily bilateral stimulation of lower extremity muscles (30 minutes per session). Proprioceptive training will be performed three days per week (25 minutes per session) consisting of body-loaded postural tasks in the horizontal position on an air bearing sled. This training aims to keep the proprioceptive and tactile systems conditioned and adaptable to upright balance challenges. Exercise training will mimic current protocols used on the ISS, but treadmill aerobic exercise will be replaced with additional cycling aerobic exercise. Primary outcome measures will include functional tests that are representative of high priority exploration mission tasks and require high demand for dynamic control of postural stability. Additional measures will be used to identify the key physiological factors contributing to countermeasure benefits, including assessments of sensorimotor function, somatosensory feedback, musculoskeletal function, muscle structure using MRI, ultrasound-guided muscle thickness and echointensity, electrical impedance myography, and DXA bone scans. Additionally, serological measurements, neuromuscular biomarker and circulating miRNAs, will be investigated for their potency to be used as biomarkers of muscle degeneration and indicators of treatment response. Associations between these changes in key physiological factors and changes in functional task performance will be explored. The Cardiovascular and Vision Laboratory (CVL) first identified the presence of a thrombus in the internal jugular vein (IJV) of a crewmember flying a long-duration mission to the ISS. Concurrently, the CVL team developed and implemented a scoring system to assess venous flow patterns in the IJV, documenting flow stasis in the left IJV in some, but not all, crewmembers; flow stasis represents a risk factor for thrombus development. Therefore, there would be value in determining if a spaceflight analog can be used to (1) identify if flow stasis develops that would pose a risk for thrombus formation, (2) determine whether IJV blood flow hemodynamics are related to the development of Spaceflight Associated Neuro-ocular Syndrome (SANS) outcomes, and (3) further characterize SANS findings including brain structural changes and assessment of retinal vascular leakage. The strict head-down tilt analog environment is a spaceflight analog that induces SANS in a subset of participants and facilitates the development of protocols and approaches to investigate SANS and associated physiological changes. The CVL assessed retinal vascular leakage in a previous bed rest campaign (SANS Countermeasure Bed Rest) using intravenous fluorescein angiography (IVFA) to determine whether optic disc edema in SANS results in leaky retinal vasculature. Preliminary data revealed leakage ~10 minutes after dye injection in those presenting with optic disc edema, but not in those who did not develop edema. These preliminary findings warrant further investigation into the timing of vascular leakage during IVFA imaging, and the potential consequence of “late leakage”. Therefore, additional measures have been added through a study augmentation to add data collection and analysis of outcome measures needed to assess SANS and IJV flow stasis and thrombus formation.
Rationale for HRP Directed Research:	This research is directed because it contains highly constrained research. Exploration missions must be accomplished under new constraints and fewer exercise hardware capabilities. New countermeasures are needed for these missions to replace cumulative sensorimotor stimuli and must be developed in collaboration with other NASA stakeholders. We have designed a countermeasure system to both train and assess proprioceptive function in-flight and are working the Astronaut Strength, Conditioning, and Rehabilitation (ASCR) group and members of the Human Physiology, Performance, Protection, & Operations (H-3PO) Laboratory to define the protocol. The project must be conducted in a spaceflight analog due to testing limitations on the International Space Station (ISS) and the need to determine efficacy in a controlled environment. Therefore, this research effort will require leveraging inhouse expertise including experience with head-down bed rest (HDBR) models to evaluate countermeasures. Outcomes will include functional tests previously established by Johnson Space Center (JSC) laboratories that represent critical mission tasks and have demonstrated performance decrements after both spaceflight and HDBR. It is also essential to track performance and understand the underlying factors that contribute to efficacy in order to translate the countermeasure to spaceflight operations and predict outcomes during exploration missions. The investigators of this research proposal have extensive experience conducting HDBR studies and a strong working relationship with other NASA stakeholders. Proprioceptive training is a novel approach, but a team of sensorimotor and rehabilitation experts to collaboratively define and test the countermeasure increases the likelihood of study success.

Research Impact/Earth Benefits:	The deliverable from this project will be proof-of-concept countermeasure designs for functional task performance with full assessment of efficacy in a spaceflight analog. If the countermeasures are effective, they will be translated for evaluation with the suite of operationally implemented in-flight ISS and exploration countermeasures. This will help mitigate the risk of altered sensorimotor function impacting critical mission tasks. The inclusion of various key physiological measures will also lead to a greater fundamental understanding of how proprioceptive, tactile, and muscular functions contribute to upright functional mobility. These findings may translate to ground-based interventions for various clinical populations that require balance training and fall risk mitigation. The added SANS measures will help further characterize the frequency and magnitude of vascular leakage associated with optic disc edema, venous blood flow and anatomical variability, and the brain structural changes.
Task Progress:	This past reporting period, study activities were coordinated across multiple projects/investigators sponsored by the Human Health Countermeasures (HHC) and Human Factors and Behavioral Performance (HFBP) elements. Inputs were provided to Research Operations and Integration (ROI) to finalize the integrated requirements document covering all activities tied to countermeasures and outcome measures. Any conflicts between these activities (e.g., impacts to science) were identified and mitigated. This document also includes experiments proposed by the German Aerospace Center (DLR) investigators. Additional documents were finalized amongst the stakeholders, such as the data management plan, data collection overview, and the daily test schedule. All proprioception training and exercise countermeasure hardware were upgraded at NASA Johnson Space Center. Final modifications were made to the transportable gravity bed hardware and the tilt board training and assessment software feedback systems. Protocols were developed and training materials were generated (e.g., videos) via human-in-the-looping testing on the gravity bed and the horizontal squat device. New equipment were made for a foam obstacle course test that is well-established in the literature, including the foam obstacles, automated timing system, and polyvinyl chloride support structures. This obstacle course will be one of the primary outcome measures of this study, representing upright functional performance requiring a high degree of dynamic postural and locomotor control. All hardware were reviewed for safety by the Biomedical Research and Environmental Sciences Division at NASA Johnson Space Center. The devices were then disassembled and shipped to :envihab. The PI team traveled to Germany to reassemble the hardware and train DLR operators to conduct data collection and countermeasure training. Minor troubleshooting was required on the transportable gravity bed to maintain the cable pulley routing; however, this was repaired, and mitigation upgrades were added. The DLR team adapted one of their medical exam tables for Leg Dexterity Testing in the horizontal position. The PI team travelled to Germany to deliver the Leg Dexterity Test system and mount the device vertically at the end of the exam table. Pilot testing on the system was completed nominally, and DLR operators were trained to conduct data collection. Importantly, the device can still be dismounted and operated on the ground as it is typically used in the clinical and athletic applications. Electrical muscle stimulation (EMS) will be performed with the, FDA-cleared Playmakar® muscle stimulator. Each stimulator can stimulate two muscles simultaneously (antagonist and agonist). Since we will be stimulating 4 lower extremity muscles on each side, we will need a total of 4 muscles stimulators being used during any one session. Each muscle requires an anode and cathode (red and black) electrodes, the cathode placed over the muscle innervation zone and the anode at a distance point elsewhere along the muscle. Stimulation protocols were developed to ensure an intense yet non-painful experience that last for 30 minutes on all 8 muscles (simultaneously). Boards were developed to keep wires and devices organized during performance of the procedures. The PI team held meetings with the DLR Magnetic Resonance Imaging (MRI) team to provide the MRI sequence requirements in order to prepare for a MRI dry run. Two MRI dry runs were conducted with some improved imaging quality. The DLR MRI team ran a final dry run to confirm that the sequence could be run in the head down tilt posture. The PI team drafted an ultrasound training plan and provided the DLR ultrasound team with in-person operator training prior to the start of the HDBR campaign. The PI team has also shipped back-up image capture hardware for the ultrasound data collection. In addition, the PI team has coordinated with the off-site clinic that will conduct the IVFA data processing. The PI team developed detailed protocols for implementation of electrical impedance myography (EIM) and muscle ultrasound in each of the 8 muscles that are undergoing EMS. However, unlike EMS, which will only be performed in a single group of participants, EIM and muscle ultrasound will be performed in all groups. The PI team supported the strategic NASA acquisition of an ultrasound device for bone density measurements during the HDBR study. The device uses Radiofrequency Echographic Multi Spectrometry (REMS) technology to assess bone mineral density (BMD) at the hip and spine. Importantly, this technology is a potential candidate for spaceflight operations due to its good precision and replicability, its transportability, and the absence of ionizing radiation. The device was ordered and set up at NASA Johnson Space Center with the vendor. Then the device was shipped to :envihab, and DLR operators were trained by the PI team to conduct measurements before, during, and after HDBR. A PhD student in the JSC Neuroscience Laboratory was awarded a HRP Investigator's Workshop (IWS) Student Grant Augmentation Award (H. Gustafson). The award will support retrospective processing and analysis of posturography data from previous spaceflight and HDBR studies. The data were gathered and organized, and processing was initiated to determine whether novel outcome measures could better assess postural control abilities. In addition, multivariate regression analyses are being conducted to investigate the effects of participant demographics on postflight and post-HDBR postural control, including sex, age, height, body mass index, flight roles, experience, and flight duration. Manuscripts will be submitted to peer-reviewed journals in the next year. The NASA Institutional Review Board (IRB) reviewed and approved all protocols for this study to be conducted by DLR at :envihab. The ethics committee at DLR also approved all protocols. The PI team provided DLR with promotional materials to help recruit participants. Twelve participants (plus backups) were down-selected from hundreds of applications and screening exams for the first HDBR campaign, which began in September 2024. Two weeks of baseline data collections were completed for all twelve participants, and the first days of HDBR were initiated.
Bibliography Type:	Description: (Last Updated: 12/23/2025)

Abstracts for Journals and Proceedings	Macaulay T, Rutkove S, Peters B, Oddsson L, Wood S, Bent L, Valero-Cuevas F, Lawrence E, Strock N, Young M, Bloomberg J, Mortreux M, Bouxsein M, Beheshti A, Abitante T, Newman D. "Countermeasures for mitigation of sensorimotor decrements following head-down bed rest." 2024 NASA Human Research Program Investigators' Workshop, Galveston, Texas, February 13-16, 2024. 2024 NASA Human Research Program Investigators' Workshop, Galveston, Texas, February 13-16, 2024. , Feb-2024
Articles in Peer-reviewed Journals	Bellisle RF, Peters BT, Oddsson L, Wood SJ, Macaulay TR. "A pilot study to evaluate the relationships between supine proprioception assessments and upright functional mobility." Brain Sci. 2024 Jul 30;14(8):768. https://doi.org/10.3390/brainsci14080768 ; PubMed PMID: 39199462; PubMed Central PMCID: PMC11352215 , Jul-2024